Sahaj Bhadja

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Summary

Software engineer with 2+ years building **high-throughput systems**, **simulation-adjacent tooling**, and **real-time data workflows** using Python, Java, C++, SQL, and cloud-native CI/CD for reliable production operations.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Operating Systems, Computer Graphics, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Data Retrieval, Operating Systems

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*

- Built **production backend workflows** handling 5K+ monthly jobs with strict validation and state transitions, improving deterministic execution behavior and reducing runtime failures in asynchronous, high-volume processing pipelines.
- Optimized multi-tenant SQL data access for 500+ users by refining indexed query paths and service boundaries, reducing latency from **1.2s to 350ms** while preserving stable system throughput under load.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*

- Implemented **concurrent processing pipelines** for 10K+ daily grading requests, coordinating multimodal model execution and API orchestration while maintaining reliable response quality during peak traffic patterns.
- Developed React and Flask workflows with robust edge-case handling and automated release checks, reducing turnaround time by **60%** and improving correctness in iterative, high-frequency product updates.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*

- Created Python-based evaluation pipelines across 100K+ trajectory logs, applying structured experiment design and repeatable diagnostics to analyze system behavior under varied motion and decision scenarios.
- Collaborated across research and engineering teams to convert simulation-like findings into implementable production guidance, improving model iteration speed and reducing ambiguity in technical planning and validation.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*

- Engineered **microservice and ETL components** with Python/Java, AWS Glue, and S3, improving response times by **30%** and enabling resilient data movement across distributed, production-critical integration workflows.
- Maintained 20+ automated unit/integration suites with Jenkins, PyTest, and Mockito, improving CI pipeline efficiency by **68%** while strengthening release quality and reducing environment-specific defects by **20%**.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*

- Built **real-time REST and WebSocket modules** supporting eight operational workflows and 2M+ synchronized records, improving consistency of event-driven updates and reliability of cross-service communication patterns.
- Automated Docker and Jenkins deployment workflows to cut release time by **50%**, improving reproducibility of test-to-prod pipelines and enabling faster iteration on system behavior changes.

Skills

Languages: **C++**, **Python**, **Java**, JavaScript/TypeScript, SQL
Engineering Fundamentals: **Linear Algebra**, vectors/matrices/transforms, OOP, algorithmic optimization
Backend/Systems: Flask, FastAPI, REST APIs, microservices, asynchronous workflows, performance tuning
Cloud/Infra: AWS (S3, Glue, RDS), Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD
Tooling: Linux/Windows development, debugging, automated testing, production observability

Projects

PiXMod | *Python, Next.js, MongoDB, Transformers*

- Developed an **AI-powered image processing platform** with segmentation and semantic retrieval pipelines, processing 1K+ edited assets while optimizing end-to-end performance for dynamic visual workloads and iterative tooling updates.

CPU Scheduling Simulator | *React, JavaScript, OS Algorithms*

- Implemented FCFS, SJF, SRT, RR, priority, and MLQ scheduling with interactive Gantt visualization, reducing average turnaround by **25%** and improving debugging efficiency by **20%** across 1K+ simulated process scenarios.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built a schema-aware query generation workspace supporting 10+ database environments with validation and structured execution outputs, strengthening reliability of complex data operations and reducing manual query troubleshooting effort.